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DATE: 26/06/2023

AHB2APB Bridge Design – Project Report

AHB2APB BRIDGE

- AHB2APB Bridge is an AHB slave, providing an interface for communication between high-speed AHB (*Advanced Microcontroller Bus Architecture (AMBA) Advanced High-performance Bus*) and low-power APB (*Advanced Microcontroller Bus Architecture (AMBA) Advanced Peripheral Bus*).
- AMBA AHB is for high-performance, high clock frequency system modules. It acts as the high-performance system backbone bus and supports efficient connection of processors, on-chip memories, off-chip external memory interfaces with low-power peripheral macrocell functions.
- AMBA APB is for low-power peripherals and is optimized for minimal power consumption and reduced interface complexity to support peripheral functions. A strobe signal is used here instead of a clock.
- Blocks usually connected to the AHB bus, such as High Bandwidth External Memory, High-performance ARM Processor, DMA Bus Master and High Bandwidth on-chip RAM are called the **AHB MASTERS**.
- On the other side, peripherals at the APB bus such as UART, Timer, Keypad, etc. are called the **APB PERIPHERALS**.
- The **BRIDGE** facilitates communication between an AHB MASTER and an APB peripheral. It converts the AHB transactions to APB transactions using a protocol.
- As APB is not pipelined, wait states are added during transfers to and from the APB where AHB is required to wait for the APB.
 - **BRIDGE DESCRIPTION:**
 - The bridge unit converts system bus transfers into APB transfers and performs the following functions:
 - Latches the address in a pipeline format and holds it valid throughout the transfer.
 - Decodes the address and generates a **PSELx** (peripheral select signal)
 - Drives the data onto the APB for a write transfer
 - Drives the APB data onto the system for a read transfer.
 - Generates a timing strobe, **PENABLE** for the transfer.

TOP MODULE BLOCK DIAGRAM:

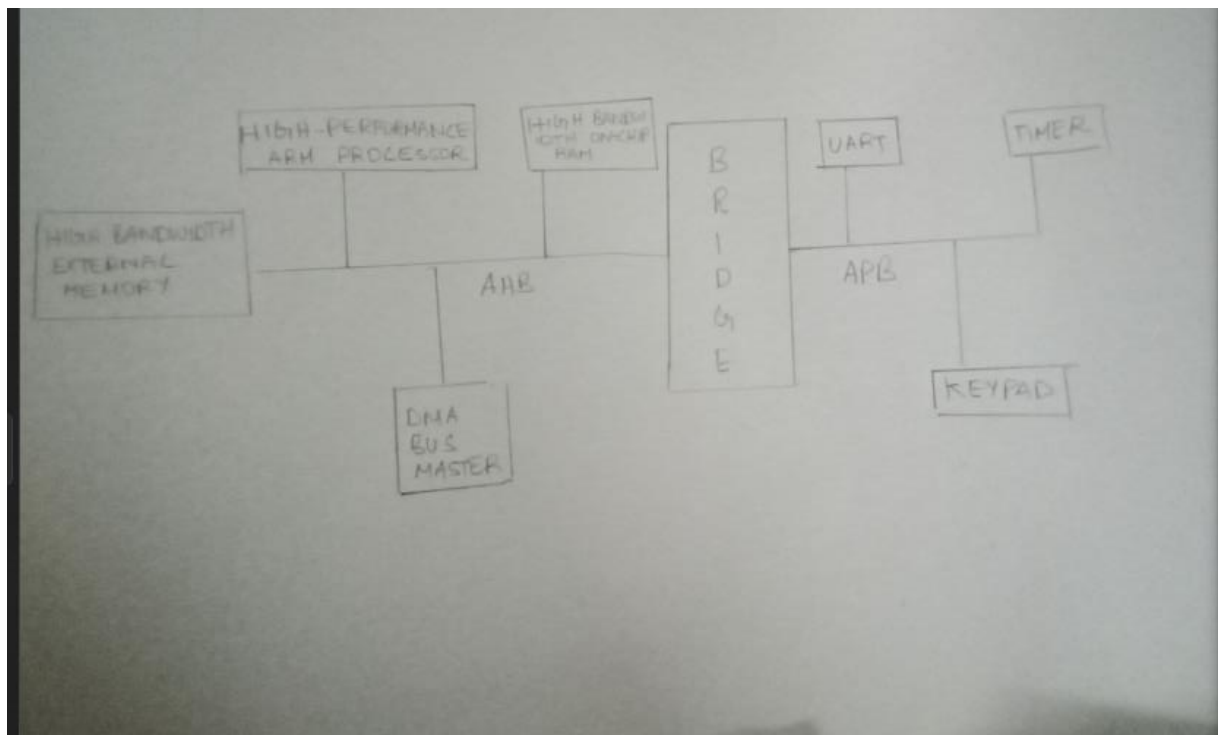


Fig 1. A typical AMBA based Microcontroller



Fig2. AHB2APB Bridge Top Module Block Diagram

AHB transactions with HREADYout signal and generation of all AOB output signal. The individual PSELx signals are decoded from HADDR using the state machine to enable the outputs while APB transaction is performed. Operation of the state machine is controlled by its current state and the AHB slave interface signals.

4. **APB Interface:** Represents the Peripheral side of the APB bus constituting the slave devices on the APB. The AMBA APB is used to interface any peripherals which are low bandwidth and do not require high-performance of a pipelined bus interface.

AHB SIGNALS:

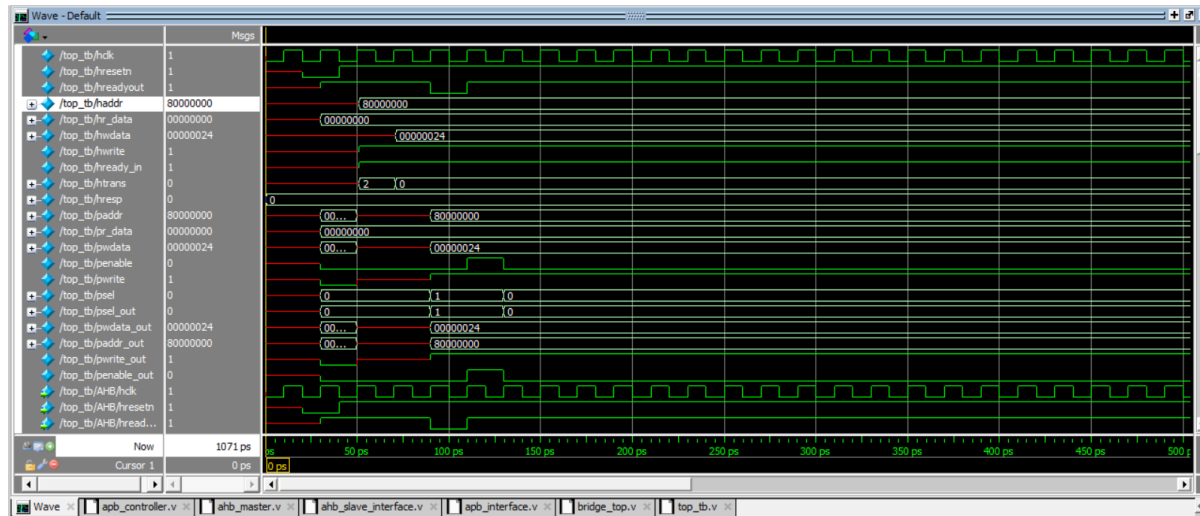
- 1) **Hclk:** system clock which is active at posedge.
- 2) **Hresetn:** active LOW reset signal to reset the system and the bus.
- 3) **Hwrite:** =0 → read, =1 → write
- 4) **Hreadyin:** indicates that the AHB master is ready for a read/write transfer
- 5) **Hwdata:** 32 bit write data bus
- 6) **Htrans:** AHB Master controls the types of transfers with the help of this 2-bit signal (IDLE - 0, BUSY - 1, SEQ - 3, NONSEQ - 2)
- 7) **Hreadyout:** signal from AHB Slave. When HIGH, indicates transfer has been finished on the bus. When LOW, can extend current transfer for multiple cycles.
- 8) **Hresp:** provides information about process of the transfer. In this bridge, Hresp is always OKAY response.
- 9) **Hrdata:** 32 bit read data bus.

PERIPHERAL SIGNALS:

- 1) **PENABLE (Peripheral Enable):** Used to time all access on the peripheral bus.
- 2) **PWRITE:** HIGH → write to peripheral, LOW → read from a peripheral
- 3) **PSELx (3 bit):** peripheral slave select signal
- 4) **PADDR (32 bit):** Represents address of the peripheral part. Whatever address that is driven through the HADDR from AHB master is driven through the PADDR.
- 5) **PWDATA:** Peripheral Write data bus which is continuously driven by the bridge .
- 6) **PRDATA:** Read data bus carrying the read data from the APB peripheral.

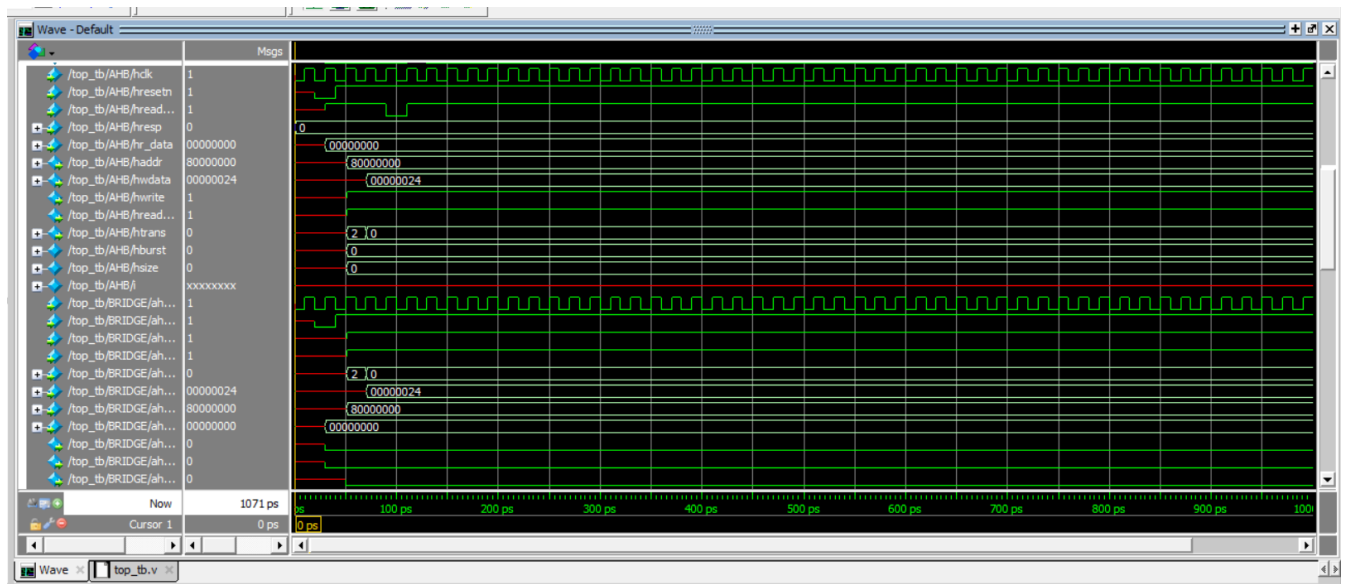
OUTPUT WAVEFORMS:

1) SINGLE WRITE:

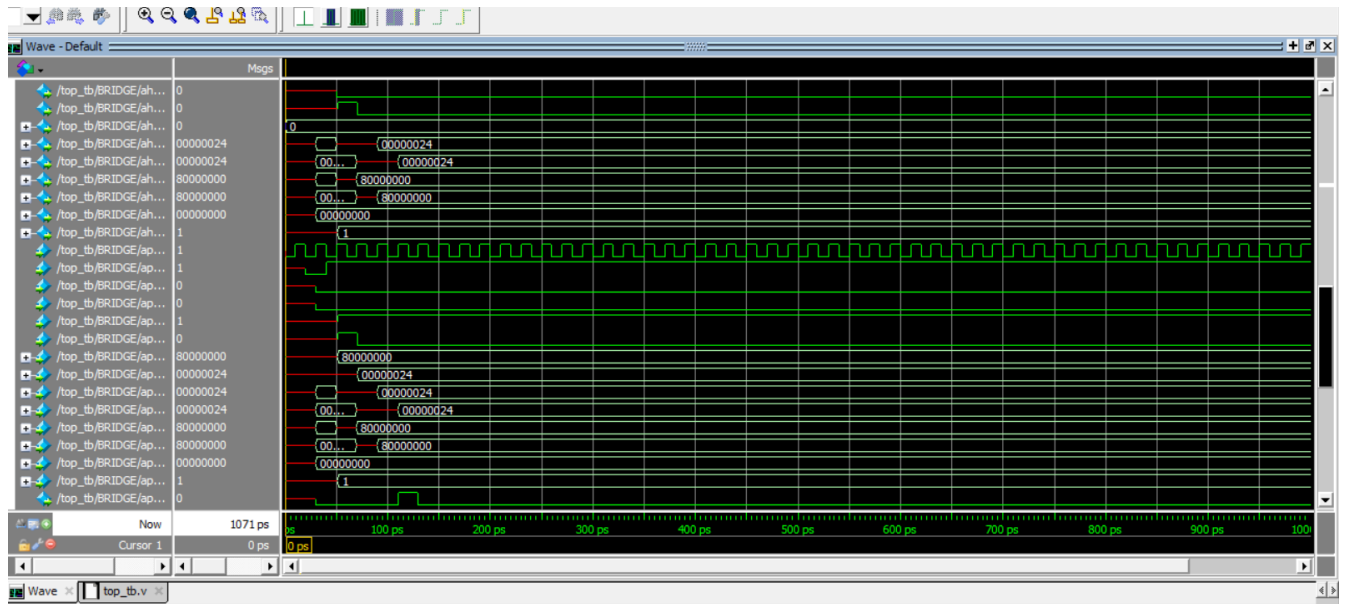


TOP MODULE OUTPUT

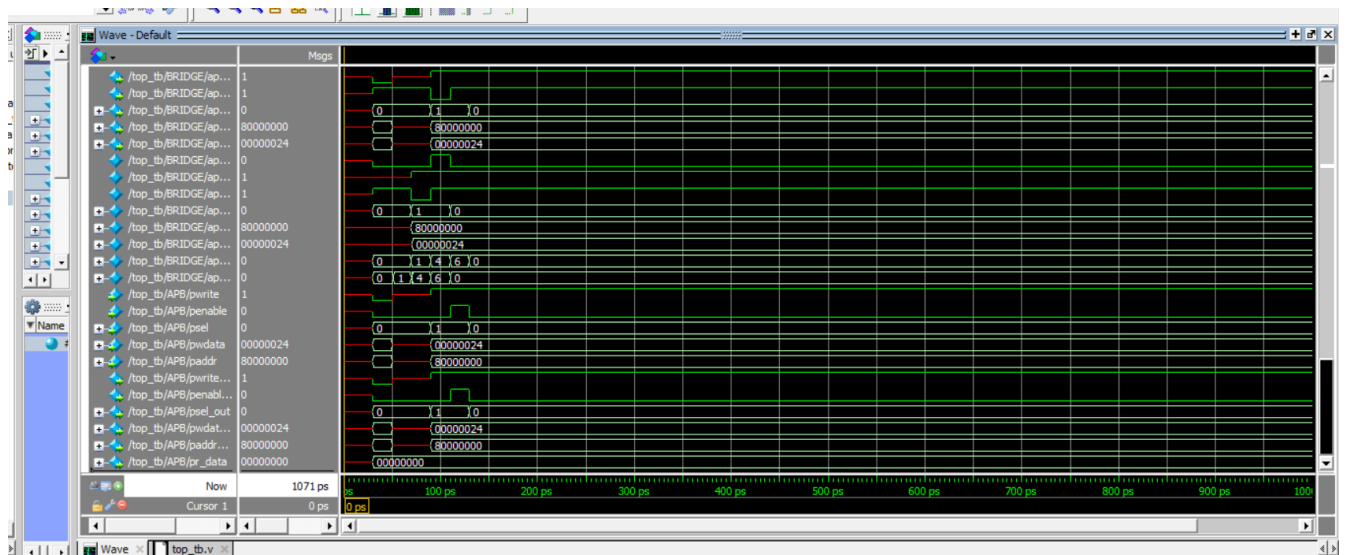
1. Address 1 (A1) driven into HADDR in first cycle when HWRITE is HIGH and HREADYOUT is HIGH
2. Data 1 (D1) in HWDATA in next cycle
3. In 3rd cycle, both A1 and D1 driven to PADDR and PWDATA respectively. PSEL set HIGH in 3rd cycle as A1 is driven to PADDR.
4. PENABLE set HIGH in 4th cycle to write D1



AHB MASTER & BRIDGE MODULE OUTPUT

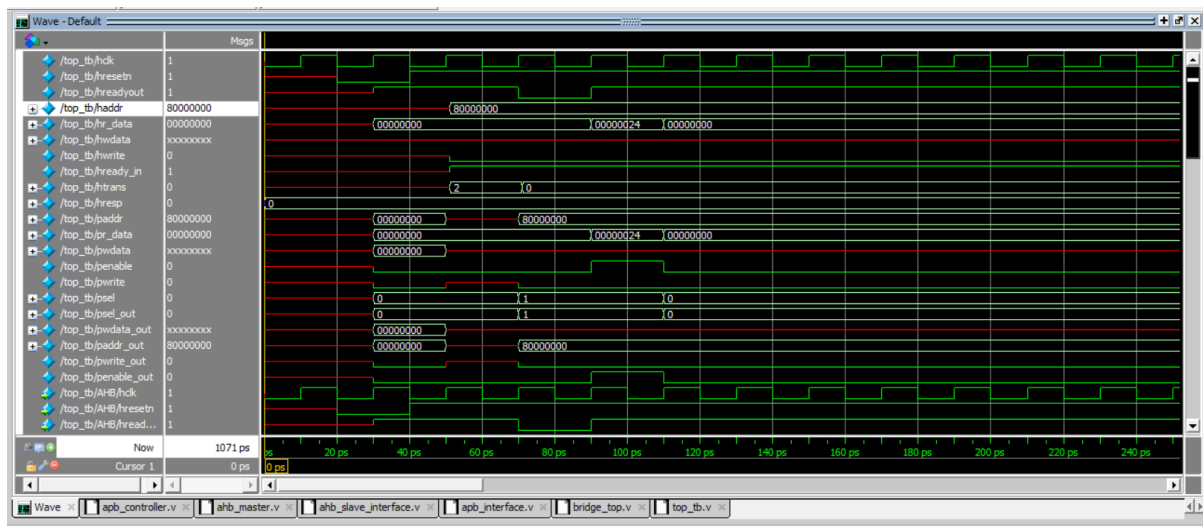


BRIDGE MODULE OUTPUT (contd.)



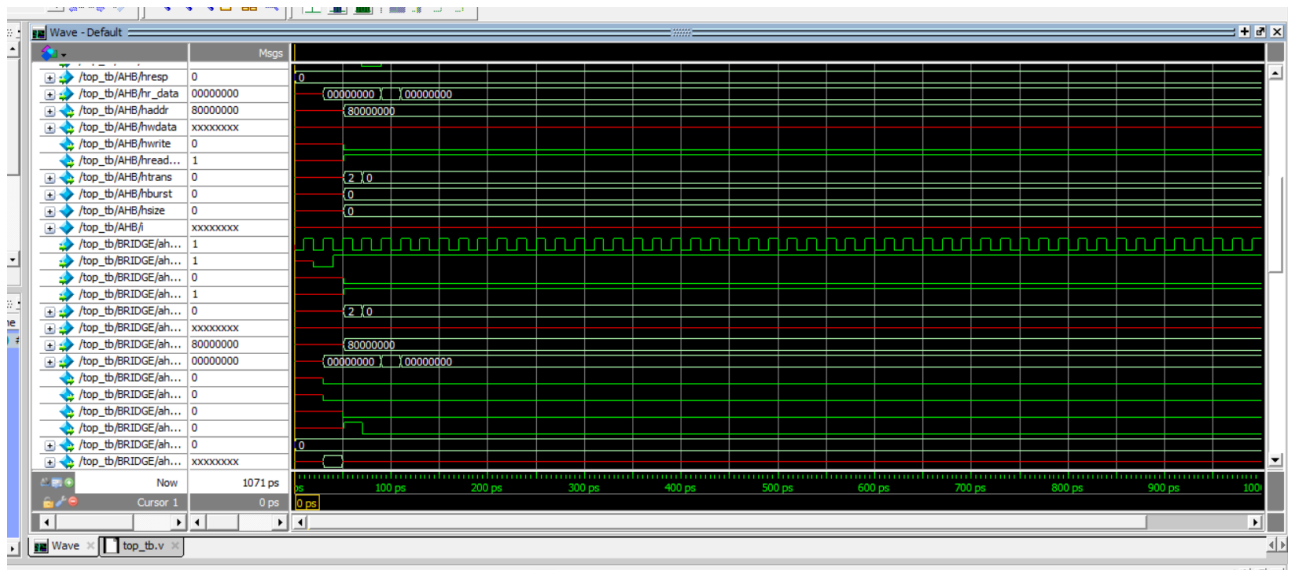
BRIDGE MODULE (contd.) & APB INTERFACE OUTPUT

2) SINGLE READ:



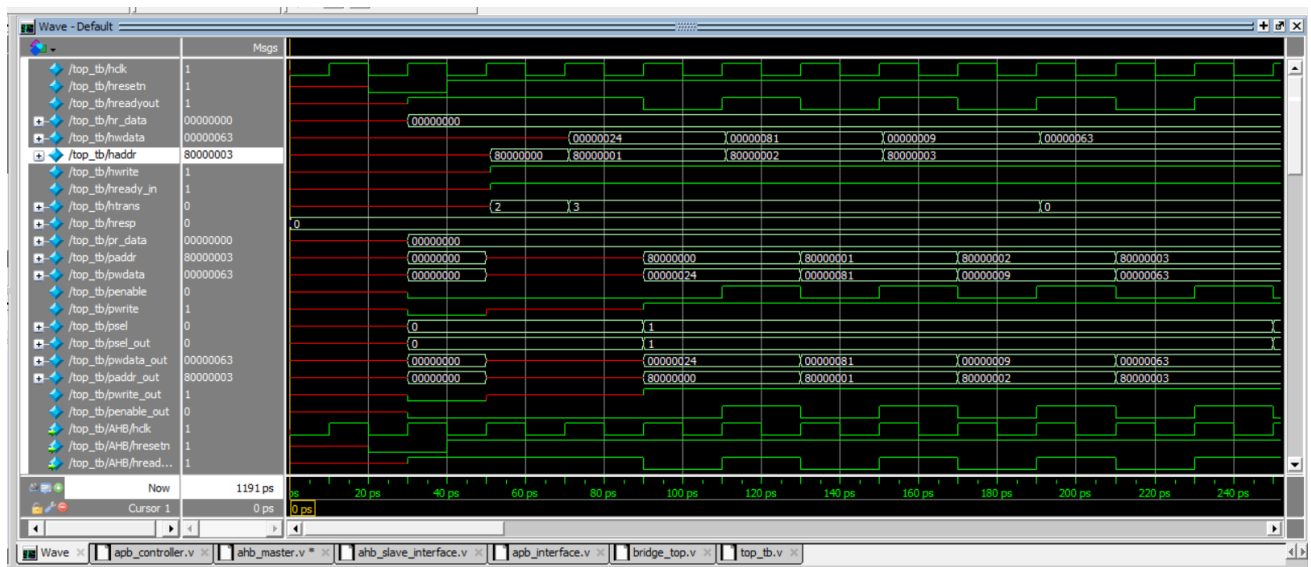
TOP MODULE OUTPUT

1. Cycle 1: HREADYOUT = 1, AHB master drives A1 into HADDR.
2. Cycle 2: A1 sampled at PADDR after which PSEL is set HIGH
3. Cycle 3: PENABLE set HIGH, data from PRDATA routed to HRDATA



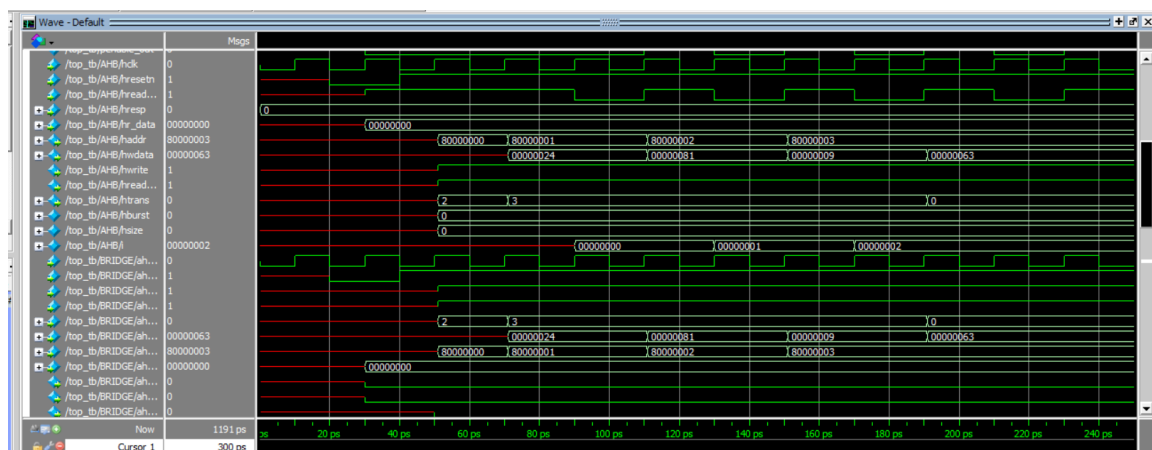
AHB MASTER & BRIDGE MODULE OUTPUT

3) BURST TRANSFER: INCREMENT 4 WRITE:

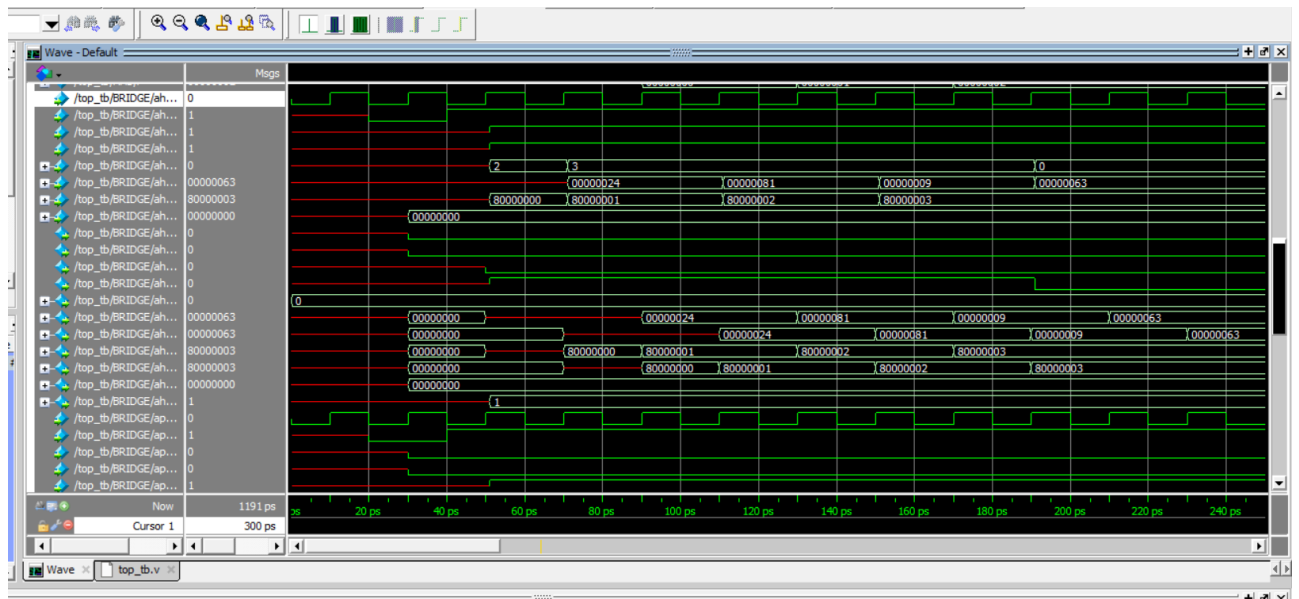


TOP MODULE OUTPUT

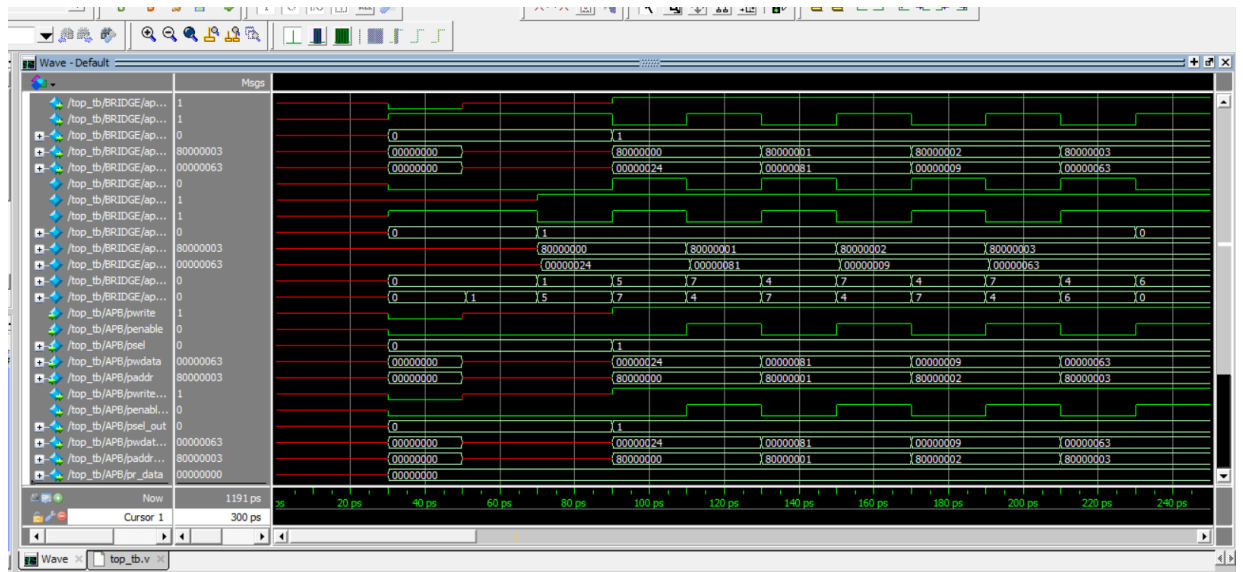
1. In 1st cycle, HREADYOUT is HIGH, A1 is driven to HADDR
2. In 2nd cycle, HREADYOUT continues to be HIGH, A2 is driven and D1 to HADDR and HWDATA respectively
3. In next cycle, A1 and D1 are sampled to PADDR and PWDATA respectively through pipelining using registers. Previous cycle, HREADYOUT was HIGH and therefore, A3 is driven to HADDR and D2 to HWDATA. But in 3rd cycle, HREADYOUT is LOW, so A3 and D2 extended till 5th cycle.
4. At 4th cycle, HREADYOUT is HIGH, so at 5th cycle, A4 and D3 driven and the process repeats.
5. PWRITE and PSEL will extend to the last transfer and PENABLE as a strobe signal will be generated four times.



AHB MASTER & BRIDGE MODULE OUTPUT

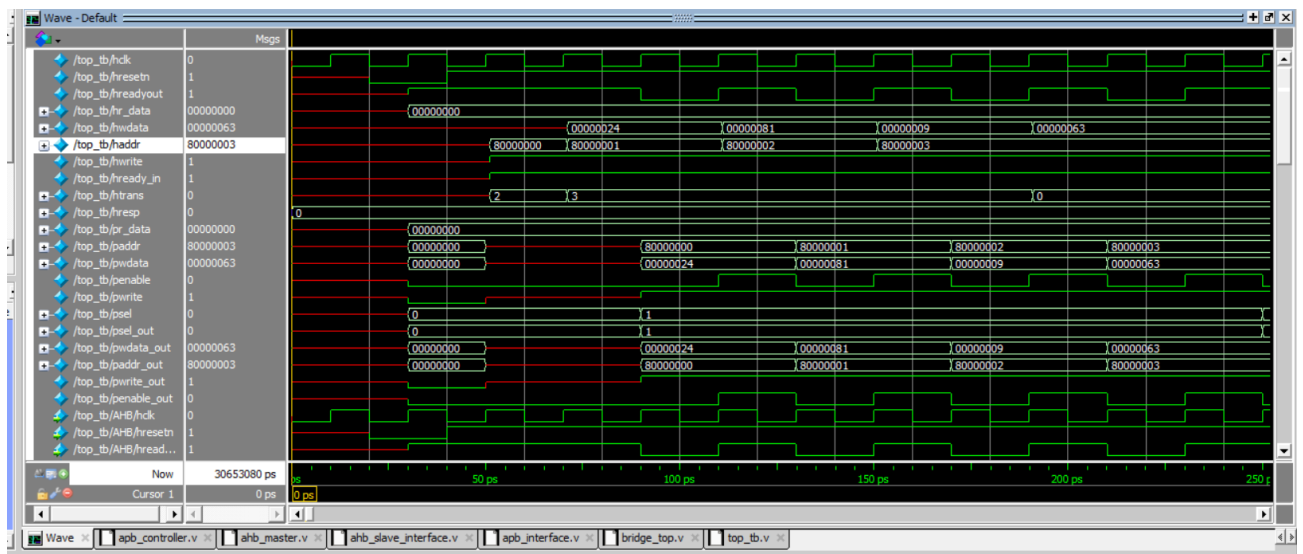


BRIDGE MODULE OUTPUT (contd.)



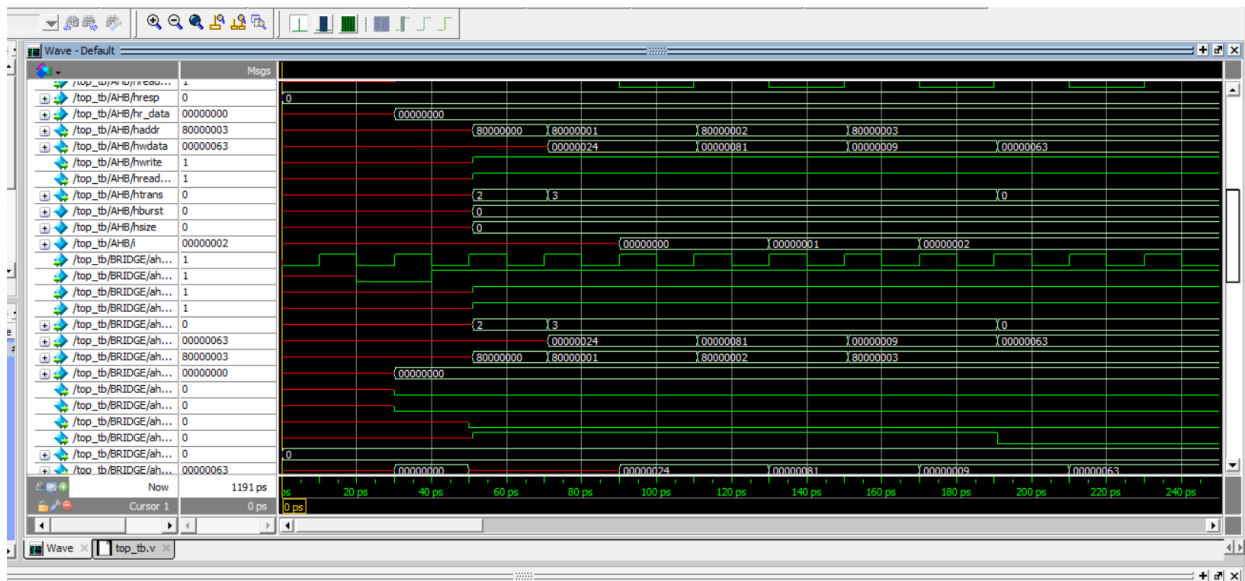
BRIDGE MODULE (contd.) & APB INTERFACE OUTPUT

4) BURST TRANSFER: WRAP 4 WRITE:

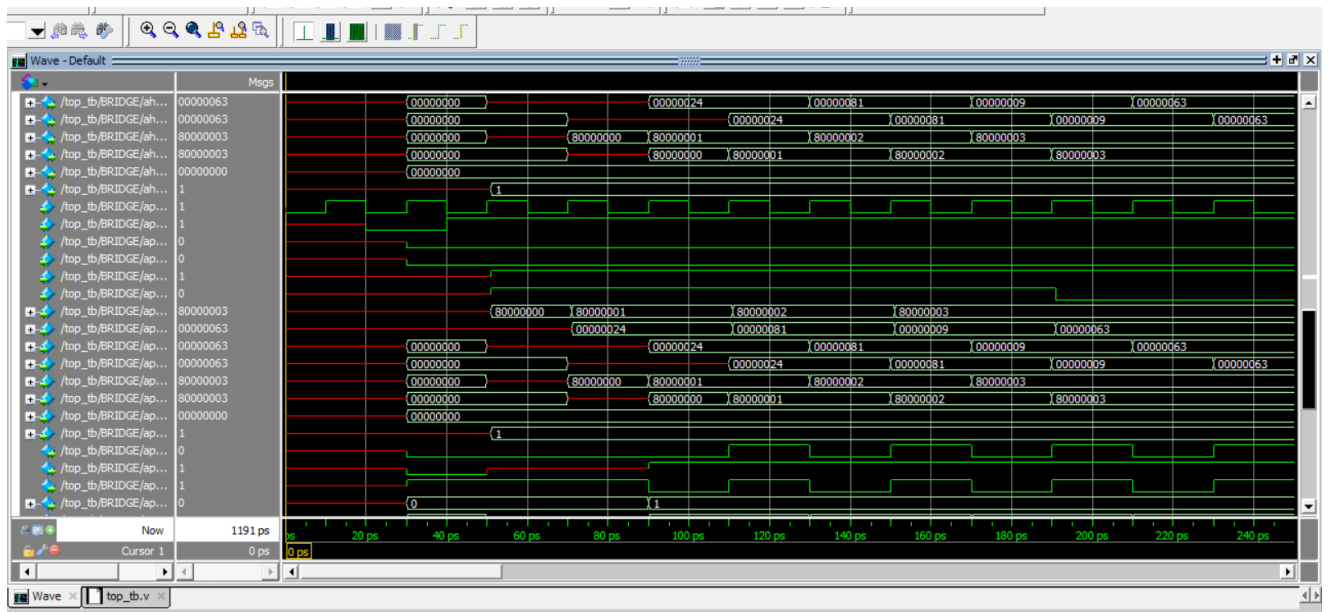


TOP MODULE OUTPUT

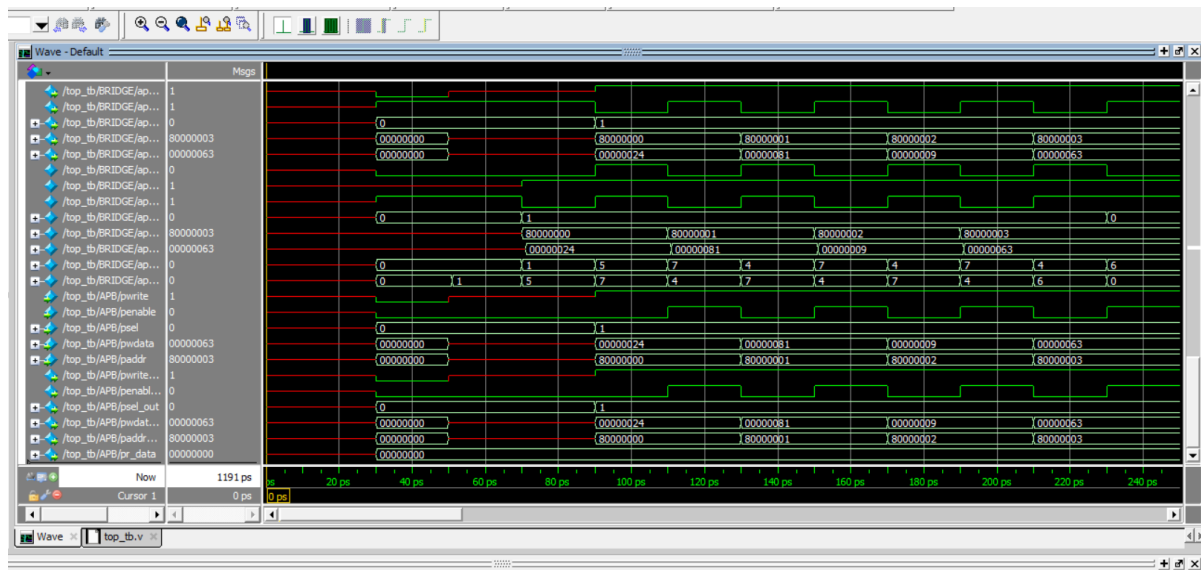
Process similar to Increment write burst, but the starting address is returned to once four increments are executed (when address boundary is reached, unlike increment burst).



AHB MASTER & BRIDGE MODULE OUTPUT

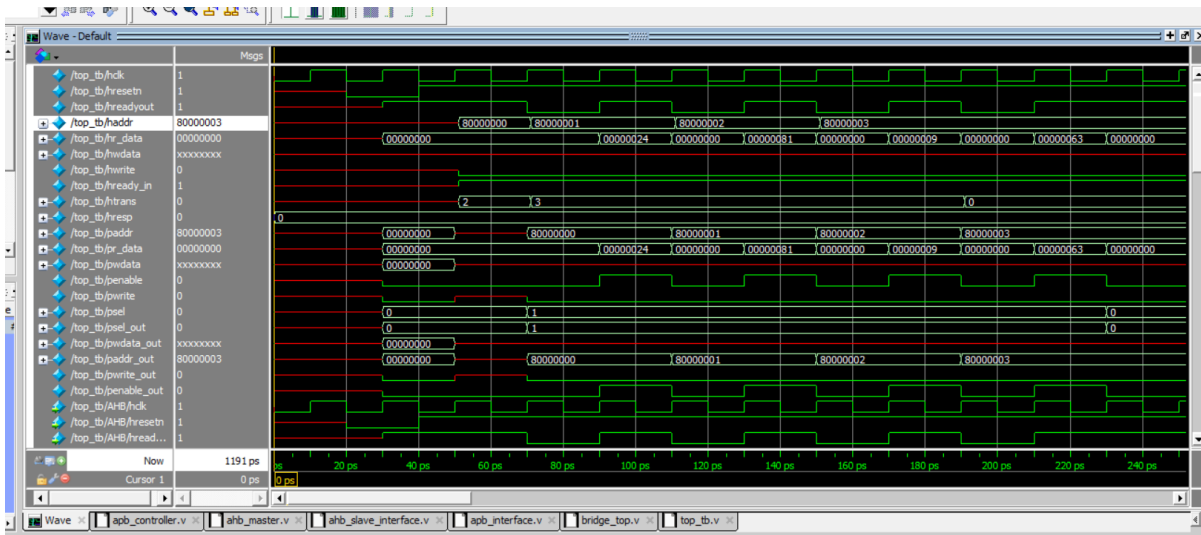


BRIDGE MODULE OUTPUT (contd.)



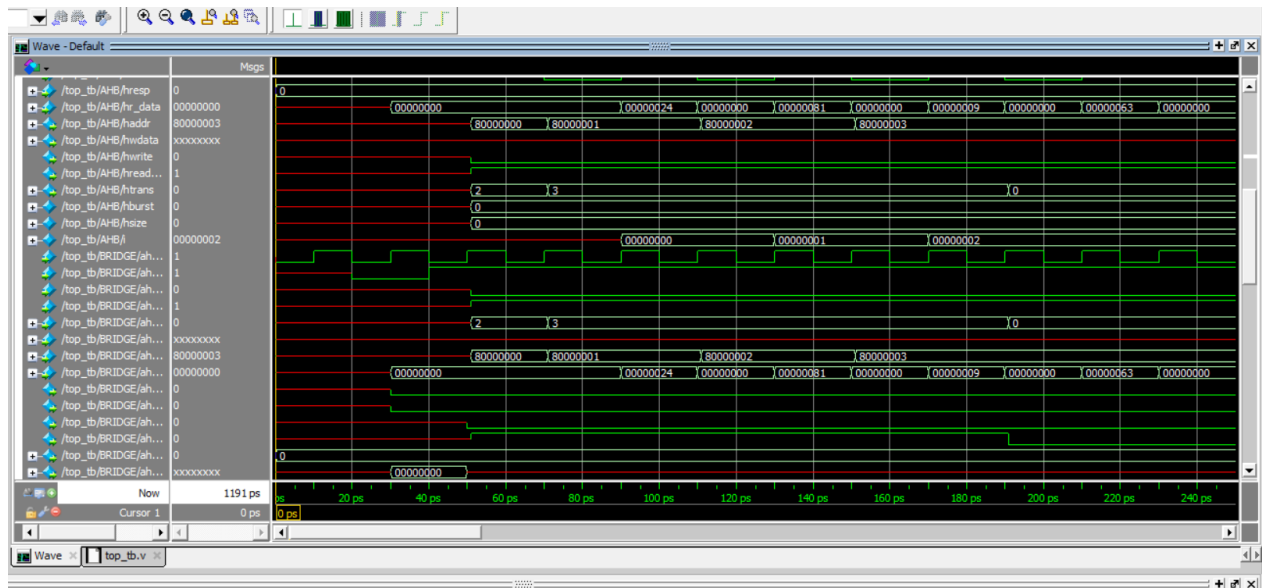
BRIDGE MODULE (contd.) & APB INTERFACE OUTPUT

5) BURST TRANSFER: INCREMENT 4 READ:

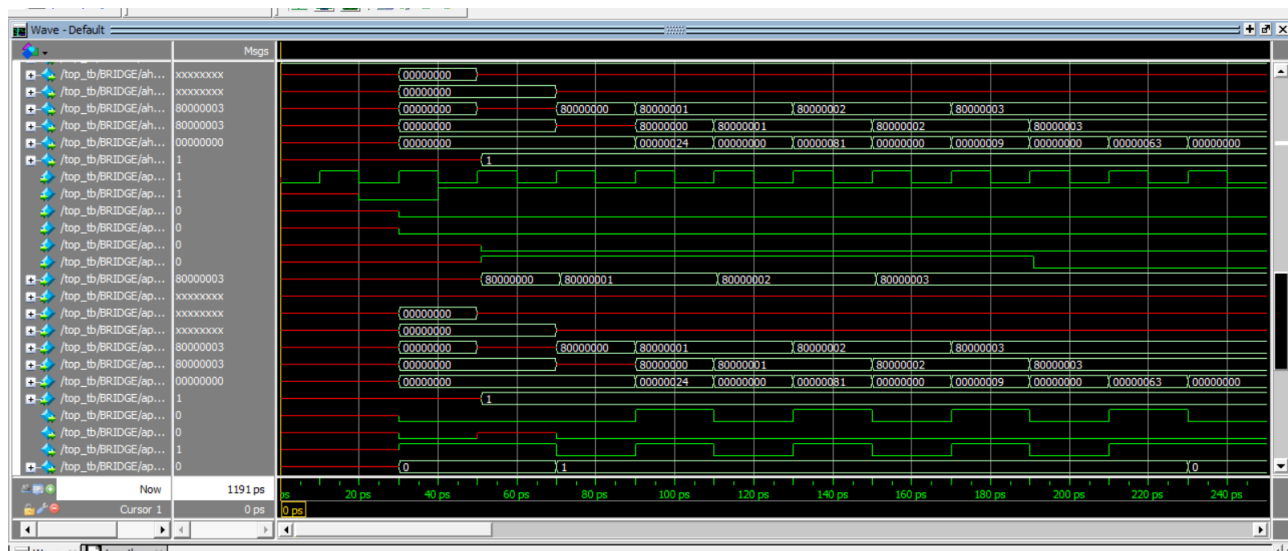


TOP MODULE OUTPUT

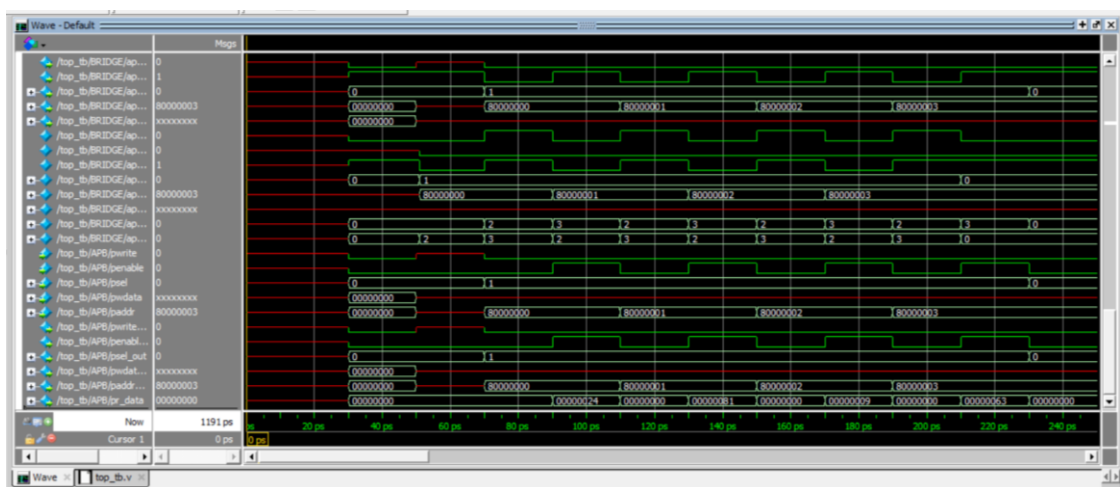
1. HREADYOUT set HIGH, A1 is driven into HADDR.
2. Next cycle, A1 driven into PADDR and PSEL set HIGH
3. Next cycle, PENABLE is HIGH, PRDATA routes D1 to HRDATA
4. At 1st cycle, HREADYOUT was HIGH and therefore A2 is driven into HADDR in 2nd cycle.
5. But since in 2nd cycle as HREADYOUT is LOW (as D1 transaction is still ongoing), A2 is extended to 2 cycles in HADDR
6. In 3rd cycle, D1 transaction ends and HREADYOUT gets HIGH again. Therefore, in 4th cycle, A3 is fed into HADDR and process repeats.



AHB MASTER & BRIDGE MODULE OUTPUT

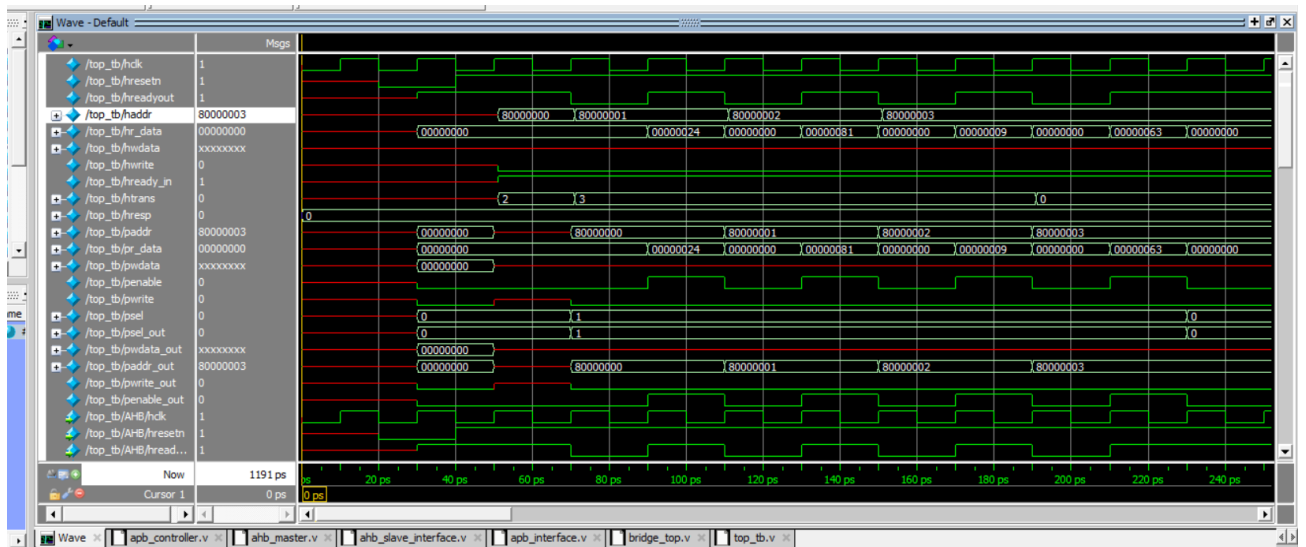


BRIDGE MODULE OUTPUT (contd.)



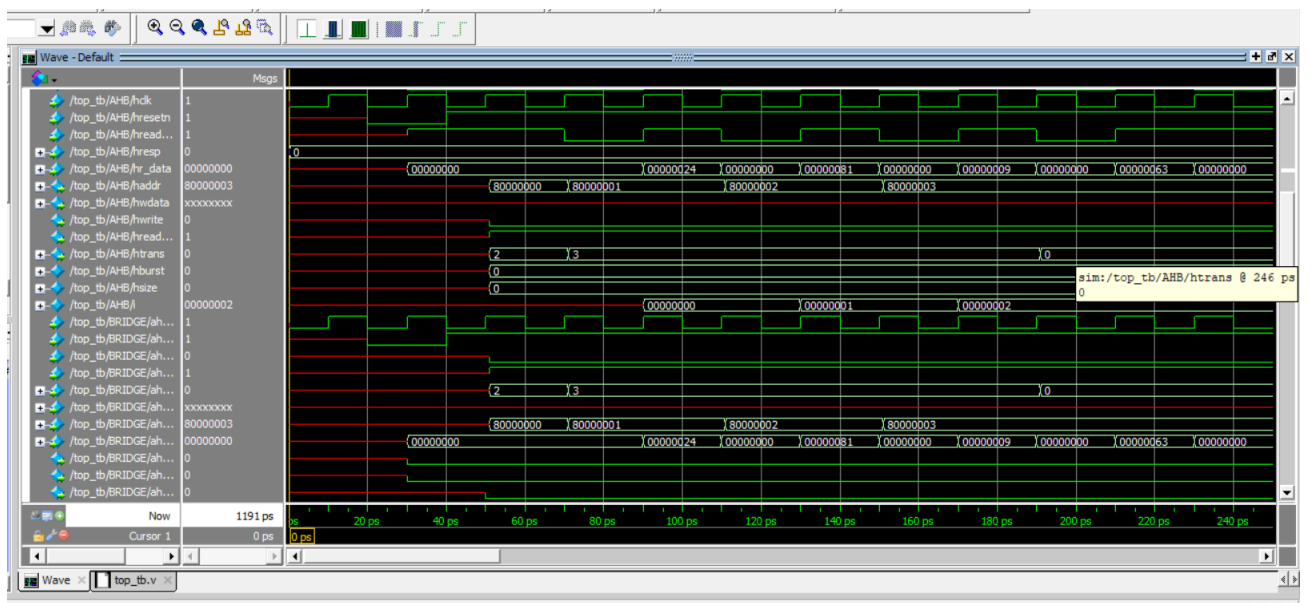
BRIDGE MODULE (contd.) & APB INTERFACE OUTPUT

6) BURST TRANSFER: WRAP 4 READ:

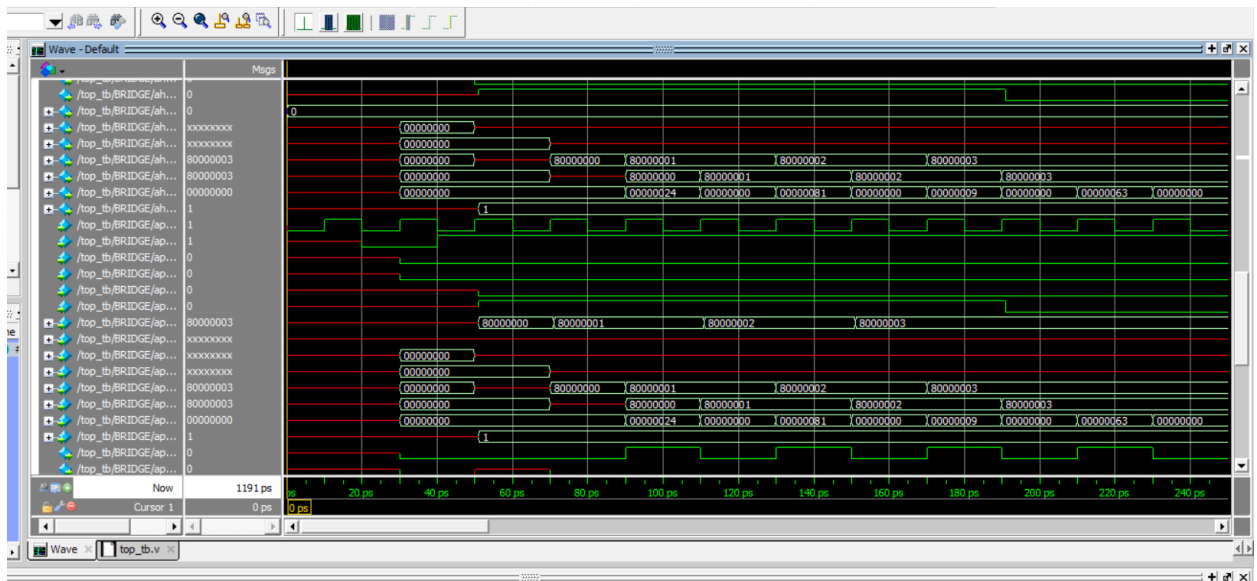


TOP MODULE OUTPUT

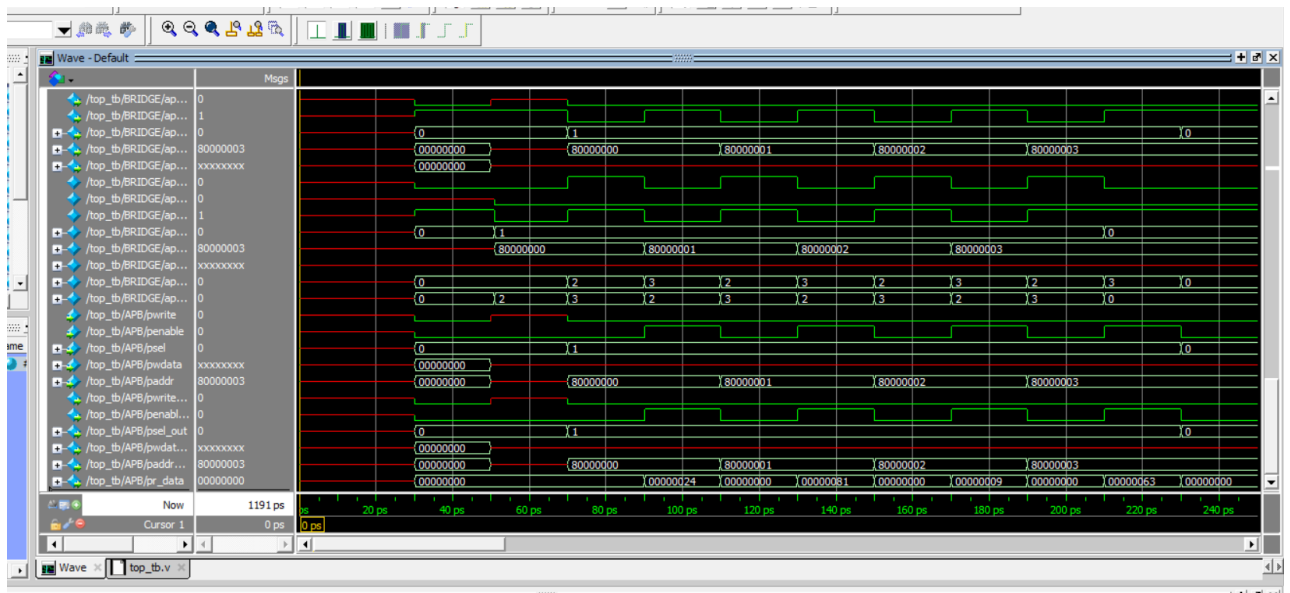
Process similar to Increment read burst, but the starting address is returned to once four increments are executed (when address boundary is reached, unlike increment burst).



AHB MASTER & BRIDGE MODULE OUTPUT



BRIDGE MODULE OUTPUT (contd.)



BRIDGE MODULE (contd.) & APB INTERFACE OUTPUT